CMSC 202 Spring 2026

Project 2 – Workout Plan Builder

Assignment: Project 2 – Workout Plan Builder

Due Date: Tuesday, March 10th at 8:59pm

Value: 80 points

1. Overview

In this project, you will:

- Practice basic C++ syntax including branching structures,
- Write classes and instantiate those classes using a constructor,
- Use arrays to hold objects,
- Use simple file input,
- Practice breaking projects into multiple files, and
- Use a makefile to compile a project.

2. Background

In this project, you will build a small console application that helps a user create a 7-day workout plan. The user can either generate a random exercise (which is one exercise per day for seven days) or choose one exercise per day for seven days.

The program loads a list of exercises from a file, then provides a simple menu to build an automatic plan or a custom plan day-by-day.

3. Assignment Description

Your program will first load exercises from an input file (exercises.txt). After loading, the program will automatically start a menu.

Using the menu, the user can view exercises, build a weekly plan, and view the current plan. (Optional extra credit adds save/load.)

4. Starting Files

For this project, you will work with the provided classes **`Exercise`**, **`WorkoutPlan`**, and **`WorkoutPlanBuilder`**. You will also use the driver file **`proj2.cpp`**.

First, navigate to your project 2 directory (for example: **~/202/projects/proj2**). Then copy the starting files into your directory.

```
cp /afs/umbc.edu/users/j/d/jdixon/pub/cs202/proj2/* .
```

Once you have run this command, you should be able to type:

```
ls
```

and see the project files listed below.

5. Requirements:

This is a list of the requirements of this application. For this project, you will be provided with header files and some starting code.

- You must follow the coding standard as defined in the CMSC 202 coding standards (found on Blackboard under course materials). This includes comments as required.
- The project must be turned in on time by the deadline listed above.
- The project must be completed in C++. Libraries we have learned include **<iostream>**, **<fstream>**, **<iomanip>**, ~~**<vector>**~~, **<cmath>**, **<ctime>**, **<cstdlib>**, and **<string>**. You may not use vectors – everything must be implemented in arrays. You should only use **namespace std**.
- You **must** use the provided header files (**Exercise.h**, **WorkoutPlan.h**, and **WorkoutPlanBuilder.h**). You may not edit any of the header files.
- No dynamic allocation. No vectors. No recursion. No global variables. No for-each loops. No auto. No EOF loops. No pointers (although pass-by-reference is ok and passing arrays is ok).
- You may not use any libraries or data structures that we have not learned in class.
- No ChatGPT or other AI-based code generation is allowed.
- No warnings or errors when you turn it in.
- Exercise names and muscle groups may contain spaces (use **getline** when appropriate).
- The number of exercises in the file is unknown but it will never exceed 100 as per the constant **MAX_EXERCISES**.

- There will always be **exactly** one exercise per day for seven days.
- If you are unfamiliar with the `const` keyword in a member function such as, `string GetName() const;` It just means the member function promises not to modify the object's data members when it runs.

6. Recommendations

Here are some general implementation recommendations:

- Read each of the header files in detail. Each header file contains important comments about what each function should do.
- Design the solution (menu flow and class interactions) on paper before you start coding.
- Stub out each function in `Exercise.cpp`, `WorkoutPlan.cpp`, and `WorkoutPlayBuilder.cpp`. Start by making sure the project compiles with the provided files (even if the most functions are not fully implemented yet).
- Implement the functions in `Exercise.cpp` as they are very simple.
- Implement the functions in `WorkoutPlan.cpp`. They are not super complicated.
- Work on one function at a time in `WorkoutPlanBuilder.cpp`. Test each feature (load exercises, randomize plan, build plan, and print plan) as you go.
- Make sure you can load exercises correctly before you attempt to build a weekly plan.
- After your menu works, test both random and custom plan building with multiple inputs.
- Your output should be as close as possible to the provided sample output. Additionally, there is a provided file named, `proj2_sample1.txt`, with more examples.

7. Sample Input and Output

The input file is `exercises.txt`. Each line contains information about a specific exercise:

Name, Muscle Group, Duration (in minutes)

```
Bicycle Crunches,Core,12
Dead Bug,Core,10
Hollow Hold,Core,9
```

Figure 1. Input File Example (exercises.txt)

In the sample output below, the user selects menu options to build and view a 7-day plan. Your exact formatting may vary slightly.

```
[[jdixon@linux3 proj2]$ make run
./proj2
Loaded 52 exercises from exercises.txt

===== Workout Plan Menu =====
1) View loaded exercises
2) Build an automatic 7-day plan
3) Build a custom 7-day plan
4) View current plan
5) Quit
Choose an option (1-5): 1
Loaded Exercises:
  1) Bicycle Crunches - Core - 12 min
  2) Dead Bug - Core - 10 min
  3) Hollow Hold - Core - 9 min
  4) Mountain Climbers - Core - 11 min
  5) Plank - Core - 10 min
  6) Russian Twists - Core - 12 min
  7) Side Plank - Core - 9 min
  8) Bird Dog - Core - 10 min
  9) Back Extensions - Back - 14 min
 10) Bent-Over Rows - Back - 16 min
```

```
11) Lat Pulldown - Back - 15 min
12) Pull Ups - Back - 14 min
13) Reverse Fly - Back - 12 min
14) Seated Row - Back - 15 min
15) Supermans - Back - 10 min
16) Bench Press - Chest - 18 min
17) Chest Fly - Chest - 14 min
18) Diamond Push Ups - Chest - 15 min
19) Incline Push Ups - Chest - 14 min
20) Push Ups - Chest - 15 min
21) Wide Push Ups - Chest - 15 min
22) Burpees - Cardio - 16 min
23) Cycling - Cardio - 20 min
24) High Knees - Cardio - 12 min
25) Jogging - Cardio - 25 min
26) Jump Rope - Cardio - 12 min
27) Jumping Jacks - Cardio - 10 min
28) Rowing - Cardio - 18 min
29) Stair Climb - Cardio - 16 min
30) Walking - Cardio - 30 min
31) Calf Raises - Legs - 10 min
32) Bodyweight Squats - Legs - 20 min
33) Goblet Squats - Legs - 18 min
34) Squats - Legs - 20 min
35) Forward Lunges - Legs - 18 min
36) Reverse Lunges - Legs - 18 min
37) Step-Ups - Legs - 15 min
38) Wall Sit - Legs - 14 min
39) Glute Bridge - Legs - 14 min
```

```
 40) Hip Thrusts - Legs - 16 min
 41) Hamstring Curls - Legs - 15 min
 42) Romanian Deadlift - Legs - 18 min
 43) Shoulder Press - Shoulders - 14 min
 44) Lateral Raises - Shoulders - 12 min
 45) Front Raises - Shoulders - 12 min
 46) Shoulder Taps - Shoulders - 10 min
 47) Arnold Press - Shoulders - 15 min
 48) Bicep Curls - Arms - 14 min
 49) Hammer Curls - Arms - 14 min
 50) Tricep Dips - Arms - 13 min
 51) Tricep Extensions - Arms - 14 min
 52) Skull Crushers - Arms - 15 min

===== Workout Plan Menu =====
1) View loaded exercises
2) Build an automatic 7-day plan
3) Build a custom 7-day plan
4) View current plan
5) Quit
Choose an option (1-5): 2

Automatic plan created.

=== Your 7-Day CMSC 202 Workout Plan ===
Monday: Cycling (Cardio, 20 min)
Tuesday: Incline Push Ups (Chest, 14 min)
Wednesday: Wall Sit (Legs, 14 min)
Thursday: Skull Crushers (Arms, 15 min)
```

```
Friday: Plank (Core, 10 min)
Saturday: Rowing (Cardio, 18 min)
Sunday: Chest Fly (Chest, 14 min)

===== Workout Plan Menu =====
1) View loaded exercises
2) Build an automatic 7-day plan
3) Build a custom 7-day plan
4) View current plan
5) Quit
Choose an option (1-5): 6
Please enter a number between 1 and 5: 0
Please enter a number between 1 and 5: 4

=== Your 7-Day CMSC 202 Workout Plan ===
Monday: Cycling (Cardio, 20 min)
Tuesday: Incline Push Ups (Chest, 14 min)
Wednesday: Wall Sit (Legs, 14 min)
Thursday: Skull Crushers (Arms, 15 min)
Friday: Plank (Core, 10 min)
Saturday: Rowing (Cardio, 18 min)
Sunday: Chest Fly (Chest, 14 min)

===== Workout Plan Menu =====
1) View loaded exercises
2) Build an automatic 7-day plan
3) Build a custom 7-day plan
4) View current plan
5) Quit
```

```
Choose an option (1-5): 3

--- Build a 7-Day Workout Plan ---
Loaded Exercises:
  1) Bicycle Crunches - Core - 12 min
  2) Dead Bug - Core - 10 min
  3) Hollow Hold - Core - 9 min
  4) Mountain Climbers - Core - 11 min
  5) Plank - Core - 10 min
  6) Russian Twists - Core - 12 min
  7) Side Plank - Core - 9 min
  8) Bird Dog - Core - 10 min
  9) Back Extensions - Back - 14 min
 10) Bent-Over Rows - Back - 16 min
 11) Lat Pulldown - Back - 15 min
 12) Pull Ups - Back - 14 min
 13) Reverse Fly - Back - 12 min
 14) Seated Row - Back - 15 min
 15) Supermans - Back - 10 min
 16) Bench Press - Chest - 18 min
 17) Chest Fly - Chest - 14 min
 18) Diamond Push Ups - Chest - 15 min
 19) Incline Push Ups - Chest - 14 min
 20) Push Ups - Chest - 15 min
 21) Wide Push Ups - Chest - 15 min
 22) Burpees - Cardio - 16 min
 23) Cycling - Cardio - 20 min
 24) High Knees - Cardio - 12 min
 25) Jogging - Cardio - 25 min
```

```
26) Jump Rope - Cardio - 12 min
27) Jumping Jacks - Cardio - 10 min
28) Rowing - Cardio - 18 min
29) Stair Climb - Cardio - 16 min
30) Walking - Cardio - 30 min
31) Calf Raises - Legs - 10 min
32) Bodyweight Squats - Legs - 20 min
33) Goblet Squats - Legs - 18 min
34) Squats - Legs - 20 min
35) Forward Lunges - Legs - 18 min
36) Reverse Lunges - Legs - 18 min
37) Step-Ups - Legs - 15 min
38) Wall Sit - Legs - 14 min
39) Glute Bridge - Legs - 14 min
40) Hip Thrusts - Legs - 16 min
41) Hamstring Curls - Legs - 15 min
42) Romanian Deadlift - Legs - 18 min
43) Shoulder Press - Shoulders - 14 min
44) Lateral Raises - Shoulders - 12 min
45) Front Raises - Shoulders - 12 min
46) Shoulder Taps - Shoulders - 10 min
47) Arnold Press - Shoulders - 15 min
48) Bicep Curls - Arms - 14 min
49) Hammer Curls - Arms - 14 min
50) Tricep Dips - Arms - 13 min
51) Tricep Extensions - Arms - 14 min
52) Skull Crushers - Arms - 15 min

Choose an exercise number for each day.
```

```
Day 1 (1-52): 10
Day 2 (1-52): 20
Day 3 (1-52): 30
Day 4 (1-52): 40
Day 5 (1-52): 50
Day 6 (1-52): 1
Day 7 (1-52): 11

Custom plan created.

=== Your 7-Day CMSC 202 Workout Plan ===
Monday: Bent-Over Rows (Back, 16 min)
Tuesday: Push Ups (Chest, 15 min)
Wednesday: Walking (Cardio, 30 min)
Thursday: Hip Thrusts (Legs, 16 min)
Friday: Tricep Dips (Arms, 13 min)
Saturday: Bicycle Crunches (Core, 12 min)
Sunday: Lat Pulldown (Back, 15 min)

===== Workout Plan Menu =====
1) View loaded exercises
2) Build an automatic 7-day plan
3) Build a custom 7-day plan
4) View current plan
5) Quit
Choose an option (1-5): 5
[jdixon@linux3 proj2]$
```

8. Compiling and Running

We have provided you with a sample makefile which should help you compile all of the classes and the program itself.

Once you have compiled using the provided makefile, enter the command make run or ./proj2 to run your program.

9. Completing your Project

When you have completed your project, you can copy it into the submission folder. You can copy your files into the submission folder as many times as you like (before the due date). We will only grade what is in your submission folder.

For this project, you should submit the following files to the proj2 subdirectory:

`Exercise.h, Exercise.cpp`

`WorkoutPlan.h, WorkoutPlan.cpp`

`WorkoutPlanBuilder.h, WorkoutPlanBuilder.cpp`

`proj2.cpp`

You do not need to submit the makefile. There is a function in the **`Makefile`** that you can use to submit. Take a look at the **`Makefile`** and see what it is called. All of the macros can be called like this, "**`make run`**" or "**`make clean`**". The submit command follows this same format.

As you should have already set up your symbolic link for this class, you can just copy your files listed above to the submission folder.

a. cd to your project 2 folder. An example might be cd
   `~/202/projects/proj2`

b. **`cp Exercise.h Exercise.cpp WorkoutPlan.h WorkoutPlan.cpp WorkoutPlanBuilder.h WorkoutPlanBuilder.cpp proj2.cpp ~/cs202proj/proj2`**

You can check to make sure that your files were successfully copied over to the submission directory by entering the command

`ls ~/cs202proj/proj2`

Make sure that the required files are submitted by the deadline. If the copy command provided does not work, it is your responsibility to figure out what is wrong and that all required files have been submitted.

You can check that your program compiles and runs in the **`proj2`** directory, but please clean up any **`.o`** and executable files. Again, do not develop your code in this directory and you should not have the only copy of your program here. Uploading of any **`.gch`** files will result in a severe penalty.

IMPORTANT: If you want to submit the project late (after the due date), you will need to copy your files to the appropriate late folder. If you can no longer copy the files into the proj2 folder, it is because the due date has passed. You should be able to see your proj2 files but you can no longer edit or copy the files in to your proj2 folder. (They will be read only)

- If it is 0-24 hours late, copy your files to **`~/cs202proj/proj2-late1`**
- If it is 24-48 hours late, copy your files to **`~/cs202proj/proj2-late2`**
- If it is after 48 hours late, it is too late to be submitted.

10. (Optional) Extra Credit

If you have completed the base project and you have additional time, you can optionally complete the extra credit feature for additional points.

STOP - Turn in your base project first. You must turn in two sets of files.

1. Base project: Turn in normally (80pts)
2. Extra Credit: Create a new directory named **`extra_credit`** and submit a second version of your project with the extra credit changes.

Extra Credit Feature: Add the ability to SAVE the current 7-day workout plan to a file and LOAD a previously saved plan from a file.

```
[jdixon@linux3 extra_credit]$ make run
./proj2
Loaded 52 exercises from exercises.txt
```

```
===== Workout Plan Menu =====
1) View loaded exercises
2) Build an automatic 7-day plan
3) Build a custom 7-day plan
4) View current plan
5) Save current plan
6) Load a plan
7) Quit
Choose an option (1-7): 4
No current plan yet. Build or load one first.

===== Workout Plan Menu =====
1) View loaded exercises
2) Build an automatic 7-day plan
3) Build a custom 7-day plan
4) View current plan
5) Save current plan
6) Load a plan
7) Quit
Choose an option (1-7): 2

Automatic plan created.

=== Your 7-Day CMSC 202 Workout Plan ===
Monday: Lateral Raises (Shoulders, 12 min)
Tuesday: Arnold Press (Shoulders, 15 min)
Wednesday: Tricep Dips (Arms, 13 min)
Thursday: Front Raises (Shoulders, 12 min)
Friday: Arnold Press (Shoulders, 15 min)
Saturday: Pull Ups (Back, 14 min)
Sunday: Russian Twists (Core, 12 min)
```

```
===== Workout Plan Menu =====
1) View loaded exercises
2) Build an automatic 7-day plan
3) Build a custom 7-day plan
4) View current plan
5) Save current plan
6) Load a plan
7) Quit
Choose an option (1-7): 5
Enter filename to save (e.g., myplan.txt): myplan.txt
Saved plan to: myplan.txt

===== Workout Plan Menu =====
1) View loaded exercises
2) Build an automatic 7-day plan
3) Build a custom 7-day plan
4) View current plan
5) Save current plan
6) Load a plan
7) Quit
Choose an option (1-7): 4

=== Your 7-Day CMSC 202 Workout Plan ===
Monday: Lateral Raises (Shoulders, 12 min)
Tuesday: Arnold Press (Shoulders, 15 min)
Wednesday: Tricep Dips (Arms, 13 min)
Thursday: Front Raises (Shoulders, 12 min)
Friday: Arnold Press (Shoulders, 15 min)
Saturday: Pull Ups (Back, 14 min)
Sunday: Russian Twists (Core, 12 min)

===== Workout Plan Menu =====
```

```
1) View loaded exercises
2) Build an automatic 7-day plan
3) Build a custom 7-day plan
4) View current plan
5) Save current plan
6) Load a plan
7) Quit
Choose an option (1-7): 2

Automatic plan created.

=== Your 7-Day CMSC 202 Workout Plan ===
Monday: Jumping Jacks (Cardio, 10 min)
Tuesday: Glute Bridge (Legs, 14 min)
Wednesday: Pull Ups (Back, 14 min)
Thursday: Push Ups (Chest, 15 min)
Friday: Chest Fly (Chest, 14 min)
Saturday: Shoulder Taps (Shoulders, 10 min)
Sunday: Step-Ups (Legs, 15 min)

===== Workout Plan Menu =====
1) View loaded exercises
2) Build an automatic 7-day plan
3) Build a custom 7-day plan
4) View current plan
5) Save current plan
6) Load a plan
7) Quit
Choose an option (1-7): 4

=== Your 7-Day CMSC 202 Workout Plan ===
Monday: Jumping Jacks (Cardio, 10 min)
```

```
Tuesday: Glute Bridge (Legs, 14 min)
Wednesday: Pull Ups (Back, 14 min)
Thursday: Push Ups (Chest, 15 min)
Friday: Chest Fly (Chest, 14 min)
Saturday: Shoulder Taps (Shoulders, 10 min)
Sunday: Step-Ups (Legs, 15 min)

===== Workout Plan Menu =====
1) View loaded exercises
2) Build an automatic 7-day plan
3) Build a custom 7-day plan
4) View current plan
5) Save current plan
6) Load a plan
7) Quit
Choose an option (1-7): 6
Enter filename to load (e.g., myplan.txt): myplan.txt
Loaded plan from: myplan.txt

=== Your 7-Day CMSC 202 Workout Plan ===
Monday: Lateral Raises (Shoulders, 12 min)
Tuesday: Arnold Press (Shoulders, 15 min)
Wednesday: Tricep Dips (Arms, 13 min)
Thursday: Front Raises (Shoulders, 12 min)
Friday: Arnold Press (Shoulders, 15 min)
Saturday: Pull Ups (Back, 14 min)
Sunday: Russian Twists (Core, 12 min)

===== Workout Plan Menu =====
1) View loaded exercises
2) Build an automatic 7-day plan
3) Build a custom 7-day plan
```

```
4) View current plan
5) Save current plan
6) Load a plan
7) Quit
Choose an option (1-7): 7
```

To be clear, the assigned version of this project is worth 80 points. The extra credit is additional and should only be attempted after the base project works.

To complete the extra credit, create a new folder in your project 2 directory named **`extra_credit`** and copy your base project files into it:

```
cp * extra_credit
```

Notice there is a space between the **`*`** and the **`extra_credit.`**

To turn in the extra credit, create an **`extra_credit`** folder in your submission folder:

```
mkdir ~/cs202proj/proj2/extra_credit
```

Then submit your **`extra_credit`** files using this command:

```
cp Exercise.h Exercise.cpp WorkoutPlan.h
  WorkoutPlan.cpp WorkoutPlanBuilder.h
  WorkoutPlanBuilder.cpp proj2.cpp
  ~/cs202proj/proj2/extra_credit
```

Final Check:

MAKE SURE THAT YOU HAVE SUBMITTED TWO VERSIONS: BASE VERSION in

```
ls ~/cs202proj/proj2
```

AND THE EXTRA CREDIT in

```
ls ~/cs202proj/proj2/extra_credit
```

You MUST turn in two completely separate versions (base version and extra credit version) of all files to be eligible for the extra credit.